ACTIVE DIRECTORY ENTERPRISE LAB

Windows Server Domain Services, Identity Management & Domain Authentication

Domain	apex.local
Domain Controller	AD-DC01
Client	Windows 10 AD Client
Platform	Oracle VirtualBox
Project Type	Hands-on Active Directory Administration Lab

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1. Executive Summary

This project demonstrates the deployment and administration of a small enterprise-style Microsoft Active Directory environment. A Windows Server domain controller was used to provide Active Directory Domain Services for the apex.local domain, while a Windows 10 client was integrated into the domain and used to validate domain authentication.

The lab focuses on core junior system administration and identity-management skills: domain services, organizational unit design, user account administration, security group management, group membership, domain joining, and authentication verification.

2. Project Objectives

- Configure and validate an Active Directory Domain Services environment.
- Use apex.local as the lab domain and AD-DC01 as the domain controller.
- Create an enterprise-style OU structure for HR, Finance, IT, and Sales.
- Create and organize domain user accounts.
- Create departmental global security groups and assign users to groups.
- Integrate a Windows 10 workstation with the Active Directory domain.
- Authenticate to the client using a domain account and verify the active identity.
- Collect clear technical evidence suitable for GitHub and interview discussion.

3. Lab Architecture

Component	Role	Evidence
AD-DC01	Windows Server domain controller running AD DS	Server Manager / AD DS
apex.local	Active Directory domain	ADUC and client verification
Windows 10 AD Client	Domain-joined workstation	systeminfo / domain login
Departmental OUs	Logical organization of identities	HR, Finance, IT, Sales
Security Groups	Role/group-based access structure	HR_Users, Finance_Users, IT_Admins, Sales_Users

4. Implementation and Evidence

Figure 1 - Active Directory Organizational Structure

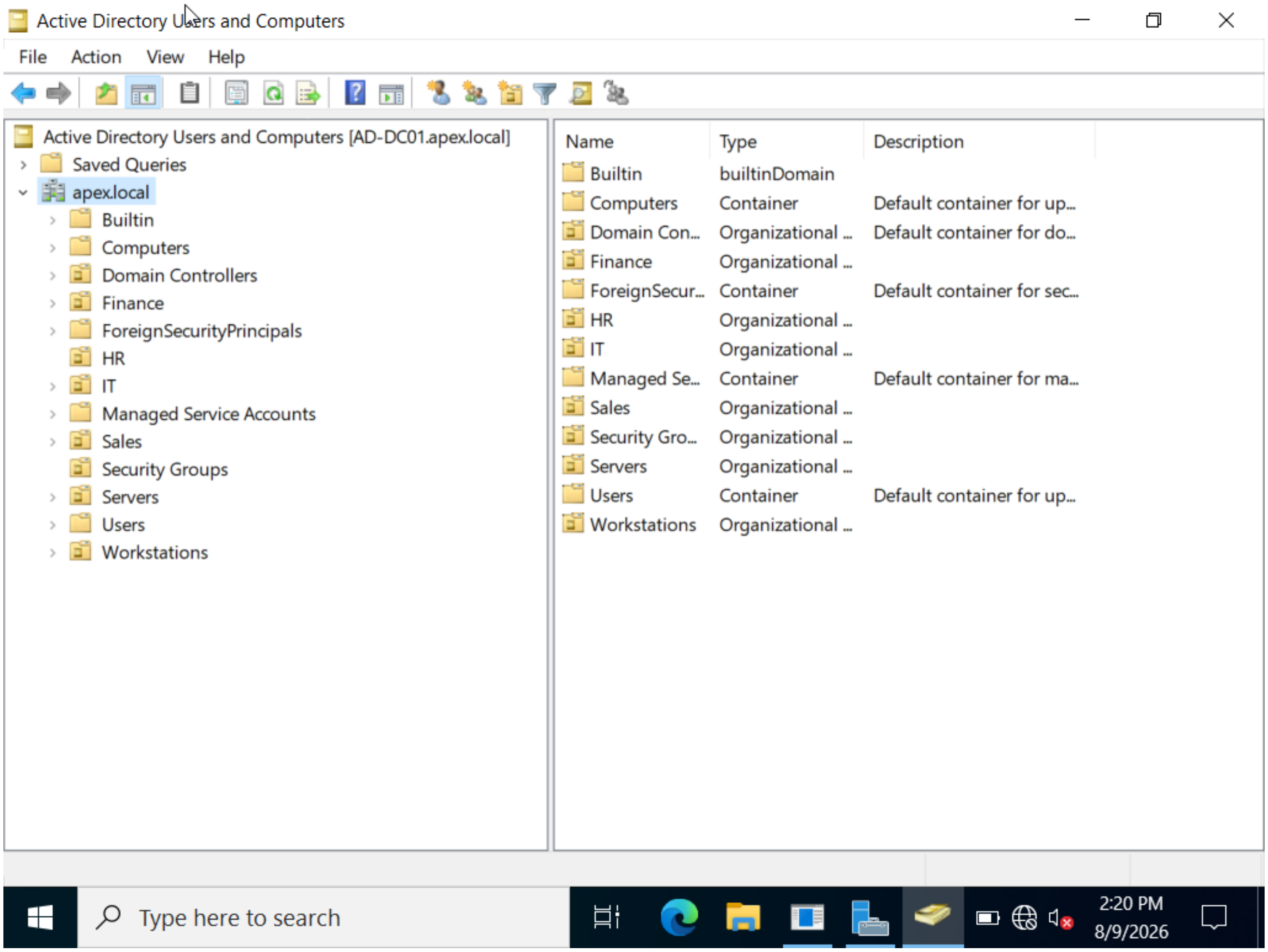

Figure 1 - Active Directory Organizational Structure

The apex.local domain was structured with departmental Organizational Units (OUs), including Finance, HR, IT, and Sales. Additional OUs were created for Security Groups, Servers, and Workstations to reflect a practical enterprise directory structure.

Figure 2 - HR Organizational Unit and User Accounts

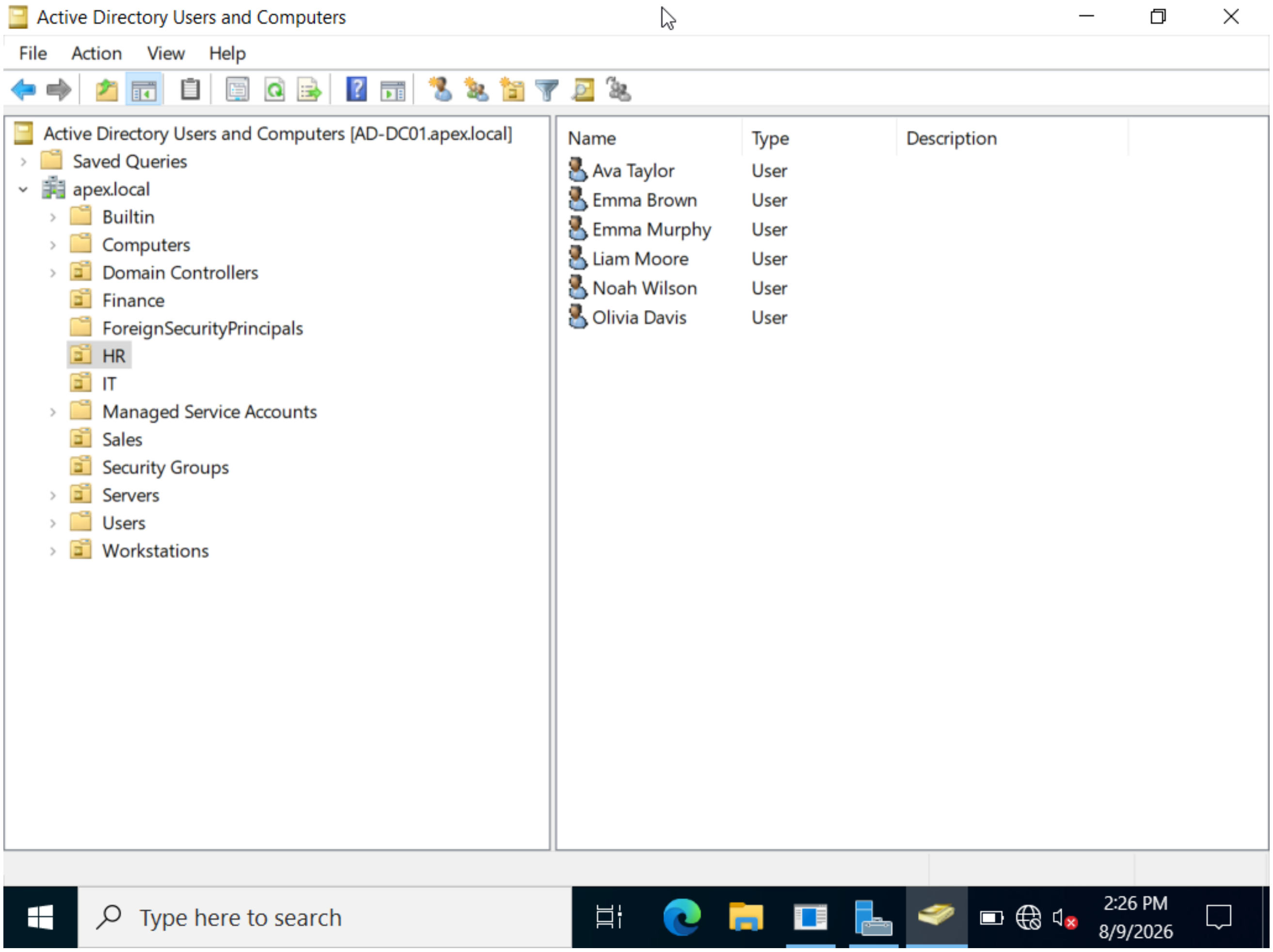

Figure 2 - HR Organizational Unit and User Accounts

The HR OU contains multiple domain user accounts, demonstrating centralized identity administration within Active Directory. The accounts include Ava Taylor, Emma Brown, Emma Murphy, Liam Moore, Noah Wilson, and Olivia Davis.

Figure 3 - Departmental Security Groups

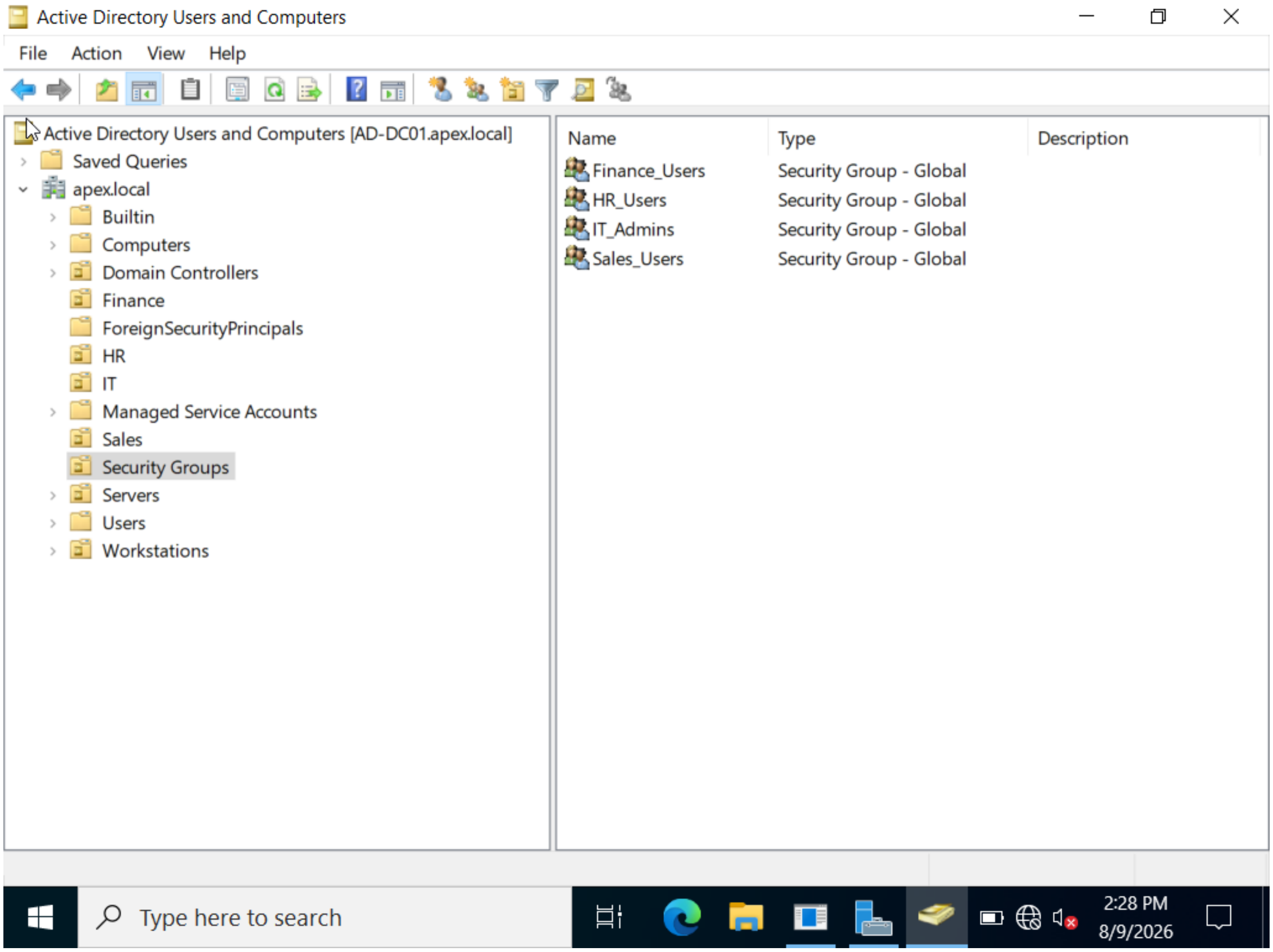

Figure 3 - Departmental Security Groups

Global security groups were created for Finance, HR, IT, and Sales. These groups provide a foundation for role-based access control and simplified permission management across enterprise resources.

Figure 4 - HR_Users Group Membership

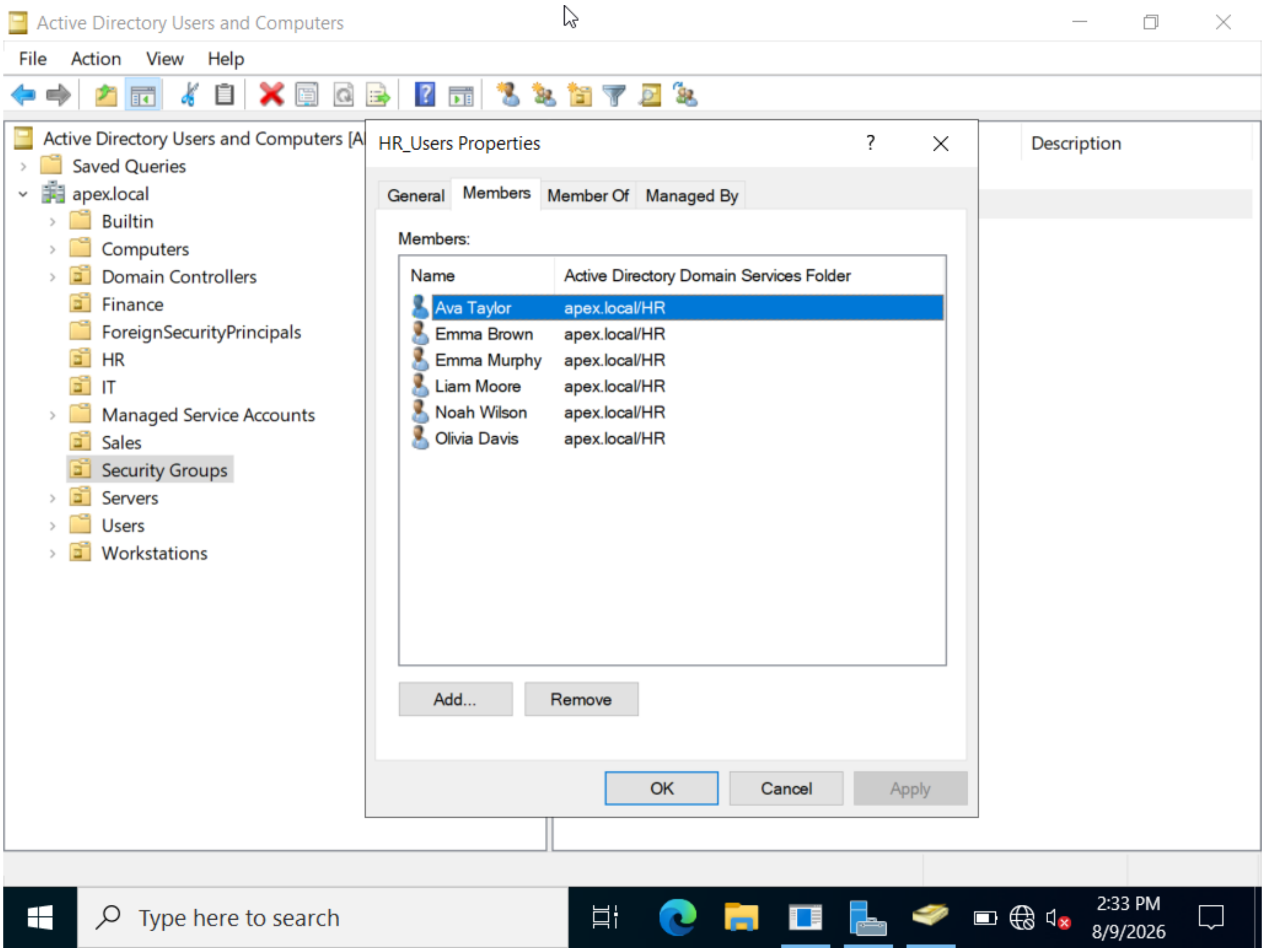

Figure 4 - HR_Users Group Membership

The HR_Users security group contains the HR user accounts, confirming that departmental users were assigned to the appropriate security group for centralized access management.

Figure 5 - Active Directory Domain Services

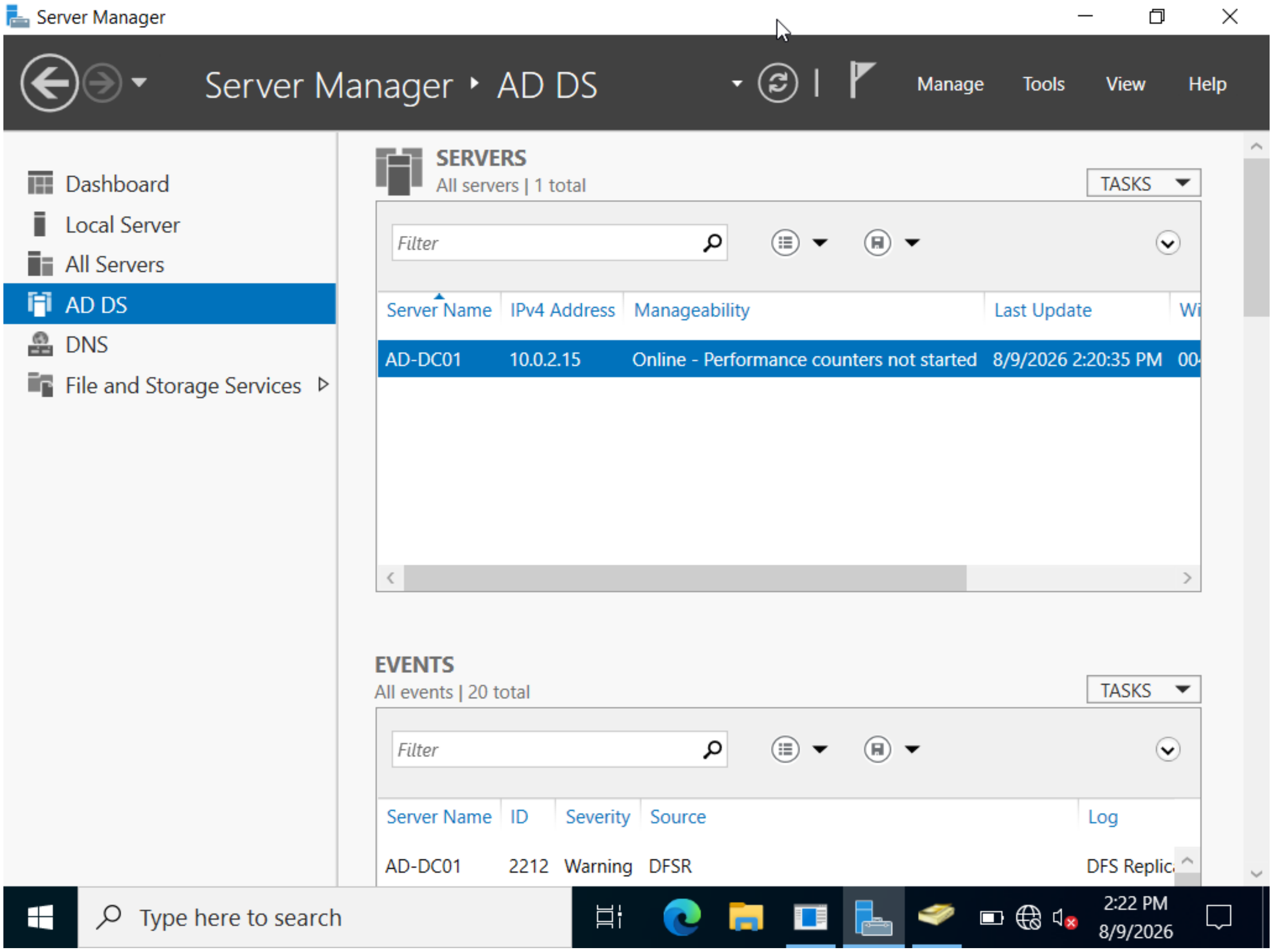

Figure 5 - Active Directory Domain Services

Server Manager confirms that Active Directory Domain Services is running on domain controller AD-DC01. The server is online and configured as the central directory services host for the lab.

5. Client Integration and Authentication Validation

Figure 6 - Domain User Authentication Verification

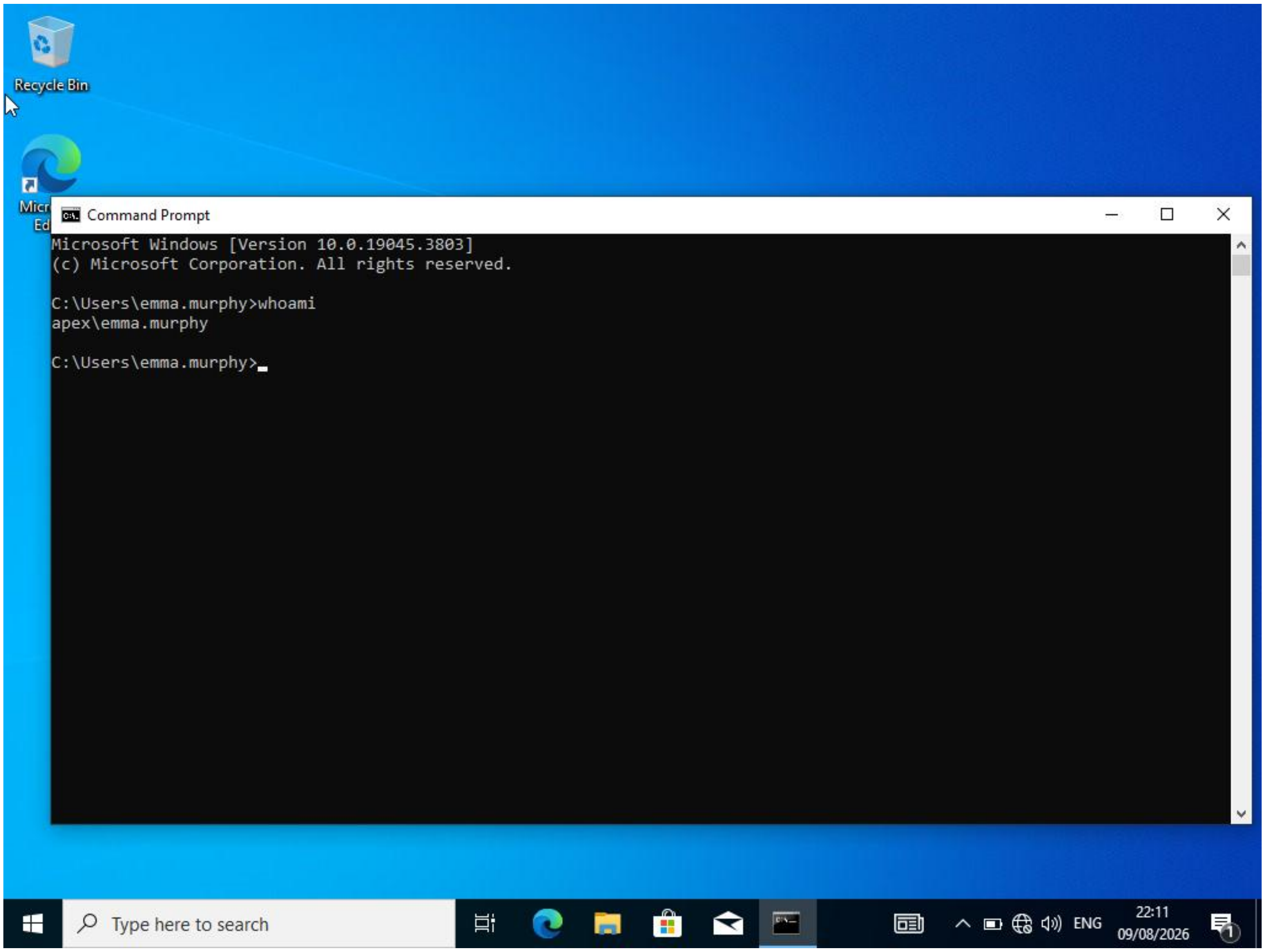

Figure 6 - Domain User Authentication Verification

The Windows 10 client successfully authenticated the domain account apex\emma.murphy. The whoami command verifies that the active session is using an apex.local domain identity rather than a local Windows account.

Figure 7 - Client Domain Membership Verification

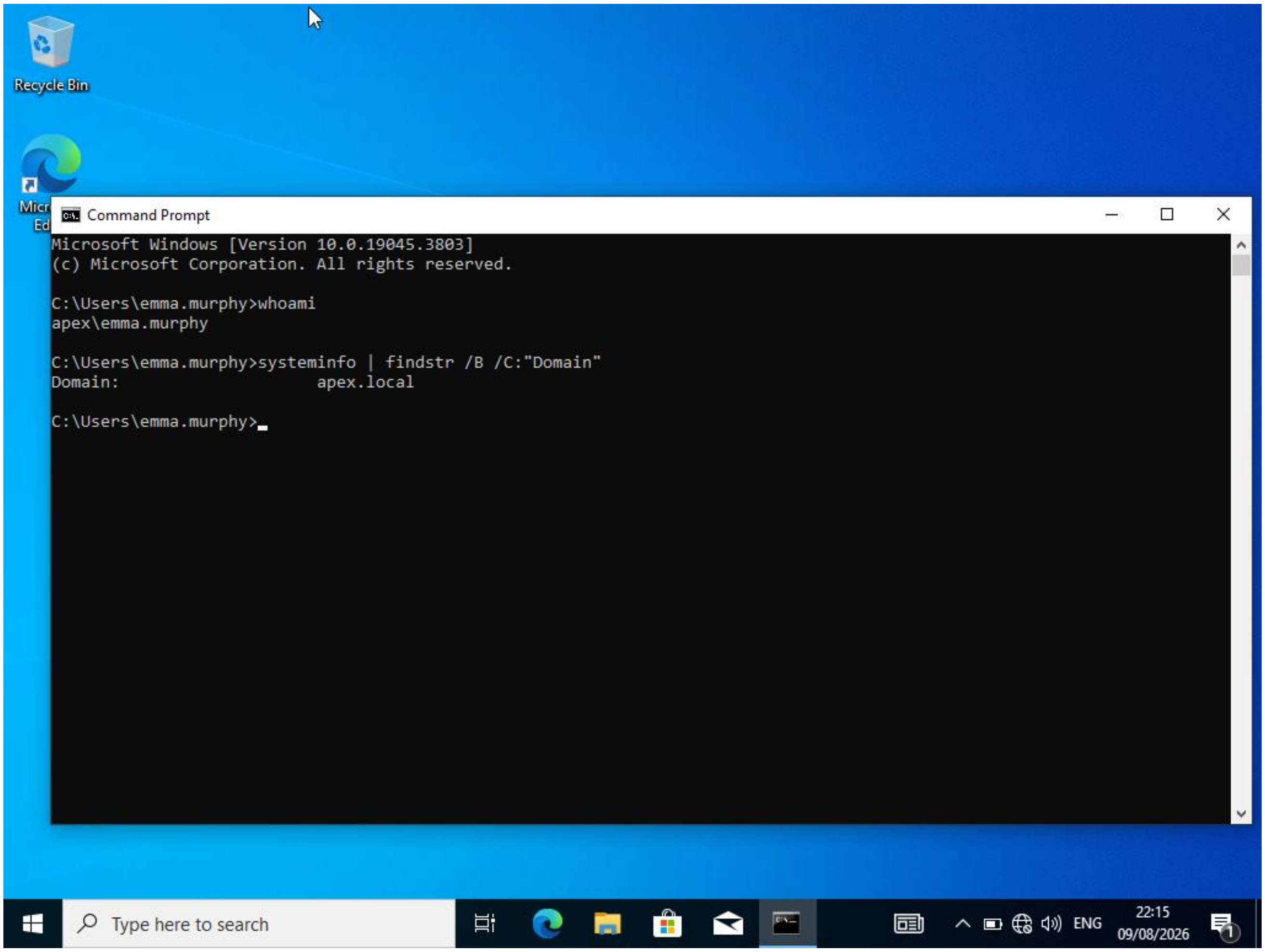

Figure 7 - Client Domain Membership Verification

The systeminfo command confirms that the Windows 10 client belongs to the apex.local domain. Together with the authenticated domain session, this validates successful client integration with Active Directory.

6. Validation Results

Validation	Status	Evidence
AD DS operational	PASS	AD-DC01 is shown online in Server Manager under AD DS.
Enterprise OU structure	PASS	Departmental and infrastructure OUs are visible under apex.local.
HR user provisioning	PASS	Six HR user accounts are visible in the HR OU.
Security groups	PASS	Finance_Users, HR_Users, IT_Admins, and Sales_Users were created as global

		security groups.
HR group assignment	PASS	HR accounts are shown as members of HR_Users.
Domain authentication	PASS	whoami returns apex\emma.murphy.
Client domain membership	PASS	systeminfo reports Domain: apex.local.

7. Skills Demonstrated

- Microsoft Active Directory Domain Services (AD DS)
- Active Directory Users and Computers (ADUC)
- Domain controller administration
- Organizational Unit (OU) design
- Domain user provisioning and account administration
- Global security group creation and membership management
- Windows domain joining and client integration
- Domain authentication troubleshooting and verification
- Command-line validation using whoami and systeminfo
- Virtualized Windows lab administration using Oracle VirtualBox

8. Security and Administration Relevance

Active Directory is a central identity and access-management technology in many Windows enterprise environments. This lab demonstrates the fundamentals behind centralized authentication, structured identity administration, and group-based access management. These concepts are directly relevant to IT administration, SOC operations, identity security, incident investigation, and access-control troubleshooting.

The departmental OU and security-group design also demonstrates the principle of managing users through logical organizational structures and group membership rather than treating every account independently.

9. Project Outcome

The lab was completed successfully. The apex.local Active Directory environment contains a functioning domain controller, departmental OUs, domain users, and global security groups. The Windows 10 client successfully authenticated as the domain user Emma Murphy, and command-line validation confirmed both the apex.local domain membership and the active apex\emma.murphy identity.The collected screenshots provide reproducible evidence of the major project milestones.